Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**
**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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**SDC HACKATHON 2026**

# Technical & Tools Declaration

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | Xeno Force |
| **Project / Solution Name** | ArcTalent |
| **Name (Leader)** | Emmanuel Heng Zi Teng |
| **Email & Phone (Leader)** | Enuelhzt1524@gmail.com 010-271-6363 |
| **University Affiliation(s)** | Universiti Malaya |
| **Live Prototype URL** | https://arc-talent--emmanuelheng.replit.app/ |
| **Code Repository URL (public / access granted to judges)** | https://github.com/MantouOP/ArcTalent |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

ArcTalent is structured as an end-to-end, privacy-first web application deployed on **Replit cloud infrastructure**:

1. **Client Tier (User Interface)**: Built with Next.js 16 and React 19, delivering an Orion dark command center, SVG constellation map, transferable skill bridge, and SME workforce heatmap. All user state and course tracking persist in **Browser LocalStorage**, guaranteeing zero personal CV data storage on the server.
2. **Backend Services (API Route Handlers)**: Next.js server routes running in Node.js handle heuristic scoring (/api/analyze), binary PDF text parsing (/api/pdf), and public market signal retrieval (/api/signals).
3. **AI Extraction Tier**: The /api/extract route dispatches confirmed synthetic CV text to a **RunPod-hosted Qwen/Qwen2.5-7B-Instruct** model via the Vercel AI SDK using structured JSON schema output. Extracted skills are strictly verified against exact verbatim quotes from the input text to prevent hallucinations, with automatic fallback to local regex keyword matching.
4. **External Signals**: Periodic public job feeds (Arbeitnow API) and research feeds (arXiv RSS) supply empirical demand weights.

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| **TypeScript (v7.0.2)** | Primary strongly-typed programming language across client and backend | Apache License 2.0 |
| **Next.js (v16.4.2)** | Full-stack web framework, App Router architecture, and serverless API endpoints | MIT License |
| **React & React-DOM (v19.3.1)** | Declarative UI rendering, reactive state management, and component trees | MIT License |

| Name & Version | Purpose in Project | License |
|---|---|---|
| **Zod (v4.6.7)** | Strict runtime schema parsing and validation for API requests and storage integrity | MIT License |
| **Lucide React (v1.61.0)** | Iconography and navigation symbols | ISC License |
| **pdf-parse (v2.4.5)** | Server-side binary PDF text stream parsing for synthetic resume uploads | MIT License |
| **Vercel AI SDK (ai v7.0.148, @ai-sdk/openai-compatible v3.0.75)** | Structured LLM communication and schema-constrained object generation | Apache License 2.0 |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| **RunPod Serverless / Proxy API** *(RunPod)* | Cloud endpoint routing inference requests to Qwen/Qwen2.5-7B-Instruct for evidence-backed CV skill candidate extraction | **Sponsor-provided** / Cloud GPU |
| **Job Board API** *(Arbeitnow)* | Live public job vacancy feed sampling market mentions for 16 transferable skills | **Free** (Public API) |
| **arXiv cs.AI RSS Feed** *(Cornell University / arXiv)* | Live artificial intelligence research paper feed providing emerging technology context | **Free** (Open Access API) |
| | | |
| | | |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| **Replit** | Primary full-stack production application hosting, Webpack build deployment, and public custom domain routing (arc-talent--emmanuelheng.replit.app) | **Sponsor-provided** / Starter Autoscale (Asia / Singapore) |
| **RunPod** | Cloud GPU container hosting the open-source LLM (Qwen/Qwen2.5-7B-Instruct) with OpenAI-compatible API serving | **Sponsor-provided** / Community Cloud (Global Proxy) |

| Service & Provider | Purpose in Project | Plan / Region |
| --- | --- | --- |
| **GitHub** | Git source code version control, continuous integration (CI) test workflows, and public repository distribution | **Sponsor-provided** / Community Cloud (Global Proxy) |
| **Client Browser LocalStorage** | Zero-backend client-side persistence for worker profiles, course completion states, and SME workforce budget settings | **Client Runtime** / Local Device |
| | | |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| **Qwen2.5-7B-Instruct** | Alibaba Cloud / Hugging Face, hosted on **RunPod GPU Cloud** | **Pre-trained** (Called via Vercel AI SDK for structured JSON extraction of skill candidates paired with exact verbatim text quotes) | Apache License 2.0 |
| **ArcTalent Rule-Based Alias Matcher** *(Fallback engine)* | Self-developed in TypeScript (src/lib/engine.ts) | **Self-developed** (Deterministic keyword/alias token matching against 16 transferable work skill definitions) | Apache License 2.0 |
| | | | |
| | | | |
| | | | |

### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| **US O*NET Content Model (v29.0)** | National Center for O*NET Development / onetcenter.org | **Public** (Standardized occupational descriptions and task requirements) | CC BY 4.0 | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| **ArcTalent Public Course Catalogue** | Curated from open public learning platforms (Coursera, edX, Kaggle, freeCodeCamp, UN SDG Academy) | **Public / Self-collected** (Public course hours, prerequisites, and URLs) | Apache 2.0 / Provider terms | N |
| **ArcTalent Synthetic Worker Cohort** *(8 fictional employee profiles)* | Project-created fixture in src/lib/catalogue.ts | **Synthetic** (Simulated workforce skills, hours, and budget for SME heatmap) | Apache 2.0 | N |
| **Synthetic Test CV** *(Aina's resume sample)* | Project test fixture (public/synthetic-cv.pdf and .txt) | **Synthetic** (Fabricated resume text used to demonstrate PDF/text skill extraction) | Apache 2.0 | N |
| **Arbeitnow Job Market Postings** | Arbeitnow Job Board API / arbeitnow.com/api/job-board-api | **Public** (Sample of European job postings used for skill-mention weighting) | Public API Terms | N |
| **arXiv cs.AI Publication Feed** | Cornell University arXiv / rss.arxiv.org/rss/cs.AI | **Public** (Metadata and abstracts of recent AI research publications) | Open Access | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No

If yes, list them below:

| Tool & Version | Used For | Extent of Use (minor / moderate / major) |
|---|---|---|
| **OpenAI Codex / ChatGPT** | Scaffolding initial Next.js architecture, heuristic matching logic (engine.ts), React UI components, and drafting documentation | **Major** |
| **Google Antigravity / Claude** | Pair-programming, debugging, running unit and production integration test suites, and refining submission materials | **Moderate** |
| **GitHub Copilot** | Inline code completion, TypeScript type annotations, and unit test case generation | **Moderate** |
| | | |
| | | |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

Our team treated all AI suggestions as draft material requiring rigorous human verification. Every algorithm and API route was vetted by team members and validated through an automated test suite comprising 32 unit tests and 14 production integration checks. We conducted manual end-to-end browser audits to ensure state consistency, privacy boundaries, and UI accessibility. All heuristic parameters (scoring weights, skill requirements, course durations) were reviewed, cross-referenced against O*NET and Malaysian workforce benchmarks, and calibrated by our team. We assume complete ownership and responsibility for the submitted codebase, architecture, and declarations.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

| Sponsor | Resource (API / credits / platform / data) | How It Was Used in the Project |
|---|---|---|
| **Replit** | Cloud platform, workspace runtime, and publishing credits | Deployed the full-stack Next.js 16 production web application, served all API route handlers, and hosts the live public URL (arc-talent--emmanuelheng.replit.app). |
| **RunPod** | Cloud GPU compute credits & serverless endpoint | Deployed a dedicated container running Qwen/Qwen2.5-7B-Instruct to provide structured GenAI skill extraction with verbatim textual evidence quotes. |
| **Aiversecity** *(Domain 1 Sponsor)* | Domain 1 problem statement, workforce challenge brief & rubric | Provided the foundational challenge scope: dynamic AI adoption tracking, transferable skills taxonomy, and alignment with UN SDG Targets 4.4, 8.2, 8.5, and 10.2. |
| | | |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Replit and RunPod provided vital cloud hosting and GPU inference infrastructure, satisfying hackathon platform criteria. However, the core system is modularly architected: the deterministic matching engine and greedy course scheduler run completely independently of external cloud services, while CV skill extraction includes an automated local regex alias matcher fallback if RunPod is offline. Replit could be replaced by standard Node.js hosting (e.g., Vercel or Docker), and RunPod by any OpenAI-compatible LLM provider. Sponsor resources significantly accelerated deployment without introducing single-point system fragility.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

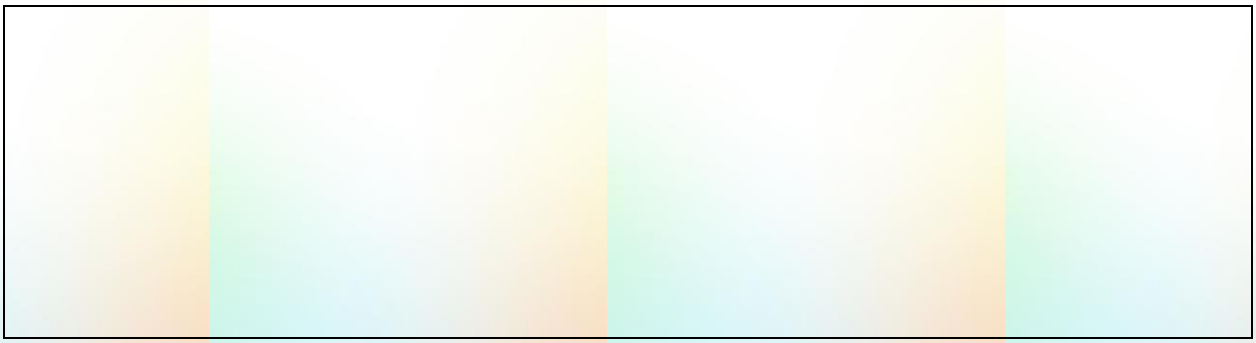
ArcTalent is engineered with a strict privacy-first, zero-retention architecture. No personal data or user accounts are stored on our servers or external databases.
All learner profile adjustments, preference choices, course completion states, and SME workforce planning parameters are stored exclusively in client-side **Browser LocalStorage**, remaining under the user's direct device control.
For CV skill extraction, our application strictly mandates synthetic test data via an explicit UI confirmation checkpoint (synthetic: true). Submitted text is processed entirely in-memory and transmitted via encrypted HTTPS to our RunPod LLM endpoint using zero-data-retention inference. CV text is never logged, stored in persistent cloud databases, or utilized for AI model retraining. Resetting the demo immediately flushes all local data from browser memory.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

 **LLM Hallucinations**: Generative models risk inferring unsupported skills.
*Mitigation*: We enforce strict programmatic verbatim substring matching (text.includes(evidence)); any candidate without an exact quote is discarded. Every extracted skill requires manual human-in-the-loop confirmation before being added to a profile.
 **Provider Dependency & Rate Limits**: Cloud GPU inference can experience downtime or throttling (enforced 8 requests/minute demo cap).
*Mitigation*: We built an instant, deterministic local regex alias-matching fallback (engine.ts) requiring zero network calls.
 **Regional Bias & External Feed Latency**: Public job feeds (Arbeitnow) represent European samples rather than Malaysian demand, and external APIs can experience network latency.
*Mitigation*: 5-minute memory caching, graceful fallbacks to retained snapshots, and an offline-capable dated synthetic replay mode.
 **Document Constraints**: The PDF parser lacks OCR and cannot process scanned image PDFs; clear user guidance prompts text-based uploads.



## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date] 25.10.2026